

AI/ML Engineer focused on building agentic AI systems using LangChain, LangGraph, and RAG pipelines that integrate LLMs with perception, reasoning, and decision-making. Experienced in designing end-to-end ML pipelines — from data processing and model benchmarking to production-scale GPU inference. Strong background in LLM APIs, prompt engineering, and iterative system design.

SKILLS

Programming	Python, SQL, C/C++, JavaScript, Bash
ML / AI	PyTorch, TensorFlow, scikit-learn, Hugging Face, NumPy, Pandas, Transformers, OpenCV
Applied AI & Systems	LangChain, LangGraph, RAG, Vector Stores, LLM APIs, Prompt Engineering, ML Pipelines, GPU Inference
Engineering Tools	Docker, Git/GitHub, Databricks, MLflow, NVIDIA Triton, Linux, GCP

EDUCATION

Master of Science in Applied Computing (CGPA 4.0/4), University of Toronto	September 2024 - December 2025
Artificial Intelligence Focus	Toronto, ON, Canada
Honors Bachelor of Science with High Distinction (CGPA 3.94/4), University of Toronto	September 2018 - April 2024
Specialist in Computer Science, Major in Mathematics, Minor in Statistics	Toronto, ON, Canada

TECHNICAL EXPERIENCE

Data Science Intern PyTorch, OpenCV, XGBoost, Databricks, Docker, NVIDIA Triton	May 2025 – December 2025
Magna New Mobility	Toronto, ON, Canada
- Built a scalable patch-based segmentation pipeline for high-resolution industrial images by splitting inputs into windows and aggregating predictions, enabling large-model inference at scale under compute constraints. - Improved detection of small and subtle defects by +0.19 IoU, +0.30 Precision, and +0.22 Recall through targeted data augmentation, architectural tuning, and optimized stitching strategies. - Developed a hybrid CV–ML workflow for leather surface defects using binary segmentation → defect crop extraction → XGBoost multi-class severity classification, enabling interpretable and modular severity predictions. - Designed and deployed end-to-end ML pipelines with Databricks, Docker, and NVIDIA Triton, supporting scalable data processing, GPU inference, and reproducible training and deployment workflows.	
Software Engineer Intern C++, Python, ROS2 Foxy, OpenCV, Linux, Git	May 2022 - June 2023
Magna Electronics	Brampton, ON, Canada
- Performed intrinsic and extrinsic camera calibration to improve perception accuracy in vision-based sensing systems, ensuring reliable sensor alignment for downstream autonomy tasks. - Strengthened robustness of autonomous sensing software by debugging and extending C++/Python modules, improving system stability under real-world operating conditions. - Supported production readiness by managing version-controlled customer deployments, while separately integrating new ROS2 Foxy C++ components on Linux as part of the platform migration effort, maintaining code quality through testing, reviews, and GitHub workflows.	

PROJECTS

MT-GPT: Interactive Museum Audio Guide Language Model Prompting	September 2024 - December 2024
- Built an interactive and multilingual museum audio guide powered by GPT models, enabling personalized narration and dynamic content adaptation based on user context and preferences. - Designed a prompt-driven control framework using Persona and Chain-of-Thought prompting to guide narrative style, reasoning depth, and response structure across diverse exhibit types. - Evaluated system behavior using quantitative readability metrics and qualitative user studies, showing consistent improvements in narrative coherence, engagement, and user satisfaction under controlled prompting strategies. - Iterated on prompt designs through systematic testing and feedback analysis, treating prompts as modular components to refine agent behavior and reasoning quality.	
Artwork Restoration with Generative Image Inpainting Python, PyTorch, git	September 2024 - December 2024
- Designed and executed a controlled benchmarking pipeline for 3 generative inpainting models (CO-MOD-GAN, ICT, LaMa) on WikiArt, scaling training to 20k paintings and evaluating on a 5k-image test set with 5 degradation masks to simulate real-world damage patterns. - Analyzed model behavior using MSE, MS-SSIM, and FID, surfacing deployment-relevant trade-offs: LaMa delivered strongest pixel and structural fidelity (MSE 0.01002, MS-SSIM 0.79205), while Co-Mod-GAN better matched global data distributions (FID 112.155). - Conducted systematic failure analysis across diverse art styles, identifying robustness limits such as facial distortion, fine-detail degradation, and breakdowns under full occlusion to inform model selection and usage constraints.	